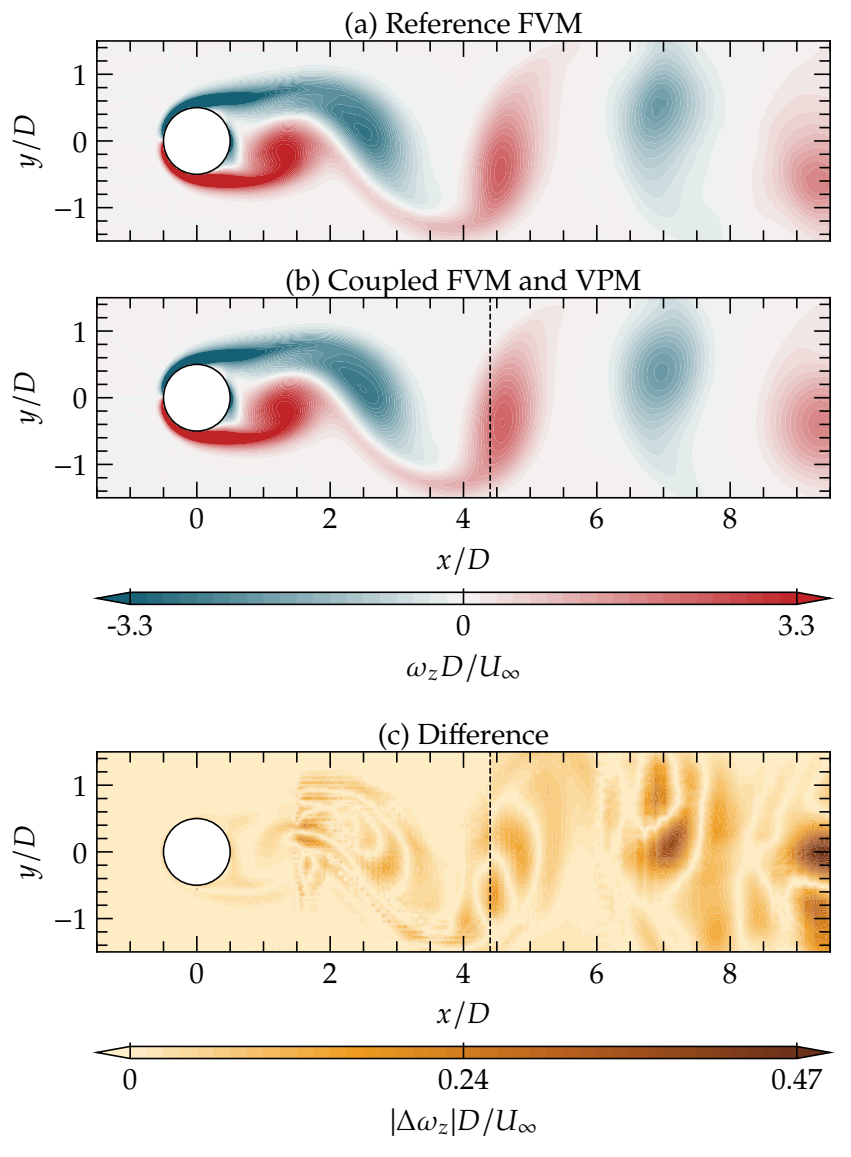

(a) Reference FVM
y/D
1
0
−1
(b) Coupled FVM and VPM
y/D
1
0
−1
0
2
4
6
8
x/D
-3.3
0
3.3
ω_z D/U_∞
(c) Difference
y/D
1
0
−1
0
2
4
6
8
x/D
0
0.24
0.47
|Δω_z|D/U_∞